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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Sixth Semester B.Tech (CBCS) Degree Examinations

May 2025

AD2206-1 – NATURAL LANGUAGE PROCESSING

(For AD)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. Which of the following is NOT a primary application of Computational Linguistics?
A) Machine Translation
B) Sentiment Analysis
C) Hardware Circuit Design
D) Speech Recognition
2. Which of the following is an example of a morphological process?
A) Tokenization
B) Lemmatization
C) Speech-to-Text Conversion
D) Dependency Parsing
3. Which is an example of collocation?
A) "Fast car"
B) "Very apple"
C) "Quick bread"
D) "Strong coffee"
4. One-hot encoding represents words as
A) Continuous vectors
B) Binary vectors with a single 1 and all other values as 0
C) Floating point numbers
D) Integer indexes
5. Which technique is used in N-gram models for predicting the next word?
A) Probabilistic estimation
B) Deep learning
C) Rule-based approach
D) Bag-of-Words model
6. TF-IDF stands for:
A) Term Frequency - Inverse Data Factor
B) Term Frequency - Inverse Document Frequency
C) Text Frequency - Inverted Document Frequency
D) Text Format - Indexed Document Factor
7. Which algorithm is known for using the "Skip-gram" and "CBOW" models?
A) BERT
B) Word2Vec
C) FastText
D) GloVe
8. What is the primary difference between traditional ML models and deep learning models for NLP?
A) Deep learning models require less data
B) Traditional ML models capture temporal dependencies better
C) Deep learning models can learn hierarchical representations
D) Deep learning models do not require training
9. Which problem do vanilla RNNs suffer from when dealing with long sequences?
A) Overfitting
B) Vanishing and exploding gradients
C) Lack of parallelization
D) Underfitting

10. What is the main limitation of the original Seq2Seq model?
 A) Struggles with long sequences due to fixed-length context vector
 B) Cannot process numerical data
 C) Too many layers
 D) Does not support parallelization
11. What is the primary difference between BERT and GPT?
 A) GPT is used for classification, while BERT is used for text generation
 B) BERT requires labeled data, while GPT does not
 C) GPT uses CNNs instead of Transformers
 D) BERT is bidirectional, while GPT is unidirectional
12. What is the main architectural component of GPT models?
 A) CNN layers
 B) LSTM cells
 C) Transformer decoder blocks
 D) Autoencoders
13. What is the main goal of topic modeling?
 A) Generate realistic text
 B) Classify documents into predefined categories
 C) Summarize long documents
 D) Discover hidden themes in a collection of documents
14. What is the goal of Text Summarization?
 A) To classify the content of the text
 B) To generate a shorter version of the text that captures the main ideas
 C) To find the named entities in the text
 D) To translate the text into another language
15. Sentiment Analysis is mainly used to determine the:
 A) Topic of a document
 B) Language of a document
 C) Emotional tone or sentiment expressed in a text
 D) Grammatical correctness of a text
16. In a Question Answering (QA) system, which approach is commonly used to retrieve the answer to a question from a large text?
 A) Information retrieval
 B) Summarization
 C) Named entity recognition
 D) Sentiment analysis
17. What is the main goal of Dependency Parsing?
 A) Find the grammatical structure of a sentence
 B) Identify named entities
 C) Break down text into smaller chunks
 D) Translate text between languages
18. Which of the following is a widely used architecture for Neural Machine Translation (NMT)?
 A) RNN
 B) Transformer
 C) CNN
 D) Naive Bayes
19. In dependency parsing, the relationship between the words is represented as:
 A) A sequence of words
 B) A list of POS tags
 C) A dictionary of named entities
 D) A tree structure
20. Which of the following evaluation metrics is commonly used to assess the performance of a classification model in NLP tasks such as sentiment analysis?
 A) BLEU score
 B) F1-score
 C) ROUGE score
 D) Perplexity

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain common challenges and problems faced in natural language processing.	8	L*2	1	1,2
	b) Explain one-hot encoding and Bag-of-Words (BoW) models. Compare their strengths and weaknesses	8	L2	2	1,2
2.	a) Define semantic ambiguity and explain how computational models can resolve it. Provide examples to support your explanation	8	L3	1	1,2
	b) What is a language model? Explain the role of n-gram models in language modeling	8	L2	2	1,2

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| 3. a) | What are collocations in the context of natural language processing? Explain their significance and methods for their extraction from text corpora. | 8 | L2 | 1 | 1,2 |
| b) | How does the skip-gram model work in neural network-based word embeddings? Explain with the suitable example. | 8 | L2 | 2 | 1,2 |

Unit – II

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|-------|---|---|----|---|-----|
| 4. a) | Explain the encoder-decoder architecture with attention mechanisms in detail. | 8 | L2 | 4 | 1,2 |
| b) | What are Conditional Random Fields (CRFs)? Explain their role in sequence labeling. | 8 | L2 | 3 | 1,2 |
| 5. a) | Explain the role of machine learning and deep learning in NLP with relevant examples. | 8 | L3 | 4 | 1,2 |
| b) | Explain BERT's architecture, pre-training tasks, and its applications in NLP tasks. | 8 | L3 | 3 | 1,2 |
| 6. a) | Compare and contrast LSTM and GRU architectures in terms of performance and efficiency with respect to NLP. | 8 | L2 | 4 | 1,2 |
| b) | Explain Graph NLP and how graph-based neural networks are used in NLP applications. | 8 | L2 | 3 | 1,2 |

Unit – III

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|-------|---|---|----|---|-----|
| 7. a) | Explain the approaches to Part-of-Speech (POS) tagging, including rule-based, statistical, and neural models. | 8 | L2 | 5 | 1,2 |
| b) | Compare extractive and abstractive text summarization techniques with examples. | 8 | L2 | 5 | 1,2 |
| 8. a) | Explain the pipeline for building a question-answering system and the challenges faced in NLP applications. | 8 | L2 | 5 | 1,2 |
| b) | Explain various visualization techniques used in NLP for model interpretation and data analysis. | 8 | L2 | 5 | 1,2 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
